STAT340 - Topic 3: Hypothesis testing

Tea-test (hypothesis testing review)

- Supposedly, R. A. Fisher invented hypothesis testing in 1935 when trying to test if British phycologist Muriel Bristol could tell if her tea was prepared "correctly", i.e. if milk is added after brewing vs before (original paper).
- Below is a slightly modified (for ease of demo) version of the original experiment:

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- Below is a slightly modified (for ease of demo) version of the original experiment:

- 8 cups are independently prepared, each with 50% chance of being done "correctly"
- After tasting, Muriel correctly guesses all 8 cups as tea-first or milk-first.

Exercises:

1. Do you think she has sufficiently demonstrated she can distinguish between tea-first and milk-first? Give a p-value and write a brief intepretation.
2. Suppose the experiment is repeated with n cups, and again Muriel guesses all n correct. What's the smallest value of n which would result in a significant test at the $\alpha = 0.05$ significance level? How about at 0.10 or 0.01?

Tea-test answers

Let $p =$ probability Muriel correctly identifies a given cup. Hypotheses:

> Null: Muriel is guessing completely at random ($p = 0.5$)
> Alt: Muriel has some ability to distinguish them ($p > 0.5$)

Note this is a binomial single proportions test from 240. Let X be number of cups correctly identified. Under null, $X \sim \text{Binomial}(8, 0.5)$. Then, p-value can be found as $\Pr[X \geq 8]$, which is just $\Pr[X = 8]$ since she guessed all 8 out of 8 correct.

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dbinom(8, size=8, prob=0.5)
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[1] 0.00390625
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The result is quite significant, so we conclude Muriel can distinguish them. Now, what's the smallest n necessary for each significance level if Muriel guesses all n correct?

```
# remember R distribution functions are all vectorized in all arguments!
dbinom(1:8, size=1:8, prob=0.5)   # alternatively: 0.5^(1:8)
```

```
[1] 0.50000000 0.25000000 0.12500000 0.06250000 0.03125000 0.01562500 0.00781250 0.00390625
```

So only 4 cups are necessary for 0.10 significance level, 5 cups for 0.05, and 7 cups for 0.01.

Original experiment

Here's the actual original experiment:

- 8 cups were prepared, 4 of them tea-first and the other 4 milk-first (this is known to the subject). Muriel was asked to identify the 4 cups that were tea-first (suppose you must pick exactly 4). Again, she correctly identified all cups. (Note this is no longer binomial!)
- This time, using an MC approach, compute a p-value for the same hypotheses.
  (hint: 1. use `sample()` to randomize cup guesses, 2. find distribution of number of correct guesses under null, 3. find p-value.) Answers (in stages):

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# 1. prepare cups and randomize guesses
cups = c(T,T,T,T,F,F,F,F)  ;  (guess = sample(cups))
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[1]  TRUE  TRUE FALSE  TRUE FALSE  TRUE FALSE FALSE
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# 2. compare guesses with cups, get total correct, repeat N=1e5 times
reps = replicate(1e5, sum(cups==sample(cups)))  ;  head(reps,n=40)
```

```
 [1] 6 4 2 8 4 6 2 6 4 2 2 4 4 2 4 4 4 4 4 8 2 2 6 2 6 4 4 4 2 4 4 4 2 2 4 4 4 4 4 4
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```
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```

```
# 3. find p-value ≈ 0.014, so again very significant
table(reps)/1e5
```

```
reps
      0       2       4       6       8
0.01429 0.22914 0.51423 0.22805 0.01429
```

Generalized testing steps

Whether using a **parametric** method (e.g. binomial test, t-test) or an MC method, the general steps for a hypothesis test are the same:

1. Identify hypotheses
    i. **Null** should be something which **predicts a specific distribution of your test statistic**, usually parameter $=$ value, or results due to random chance, etc.
    ii. **Alternative** should be opposite of null, and what you "actually want to show", e.g. Muriel can distinguish the preparation method, drug actually effective, etc.

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    - Any quantity that tends to look different under the null vs alt can be used!
    - The more "different" it tends to look, the more **statistical power** it will have.

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5. Find p-value, i.e. **appropriate tail** of distribution corresponding to your real sample data.
6. Interpret results in context of original question.

Permutation test example

A common technique useful in MC hypothesis test is the **permutation test**. Example:

- Suppose a drug was tested in a randomized controlled trial, and we have this data:

```
control = c(-0.1,-0.55,1.24,-0.97,-0.76,0.21,-0.27,-1.02,0.58,1.67,
            -1.07,0.17,1.45,0.34,1.15,0.18,-0.97,0.43,-1.39,-2.76)
treatment = c(0.54,0.36,0.59,-0.57,0.53,-0.78,-0.44,-0.98,1.31,0.5,
              0.57,0.49,-0.96,2.41,0.85,1.93,0.95,1.45,1.61,-1.16)
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- Suppose "bigger number better" applies here. Was the treatment statistically effective?

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```

- Suppose "bigger number better" applies here. Was the treatment statistically effective?

- Let null be "no difference in groups" and alternative be "true mean higher in treatment"
- Under null, two groups NOT different, so data can be **permuted** between groups. Ex:
  1. Permute data randomly between groups,
  2. Find distribution of difference in means under null,
  3. Use this to find p-value of sample difference.

Permutation test example (cont.)

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Answers (in stages):

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1. Permute data randomly between groups
2. Find distribution of difference in means under null,
3. Use this to find p-value of sample difference.

Answers (in stages):

```
# 1. permute data and compute a single run of mean difference
(permuted = sample(c(control,treatment)))
```

```
 [1]  1.45  0.53  0.50  2.41  1.24  1.67 -1.16  1.45  1.61  0.34  1.93 -0.78  1.15 -0.96 -1.39 -0.98
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```

```
mean(permuted[1:20]) - mean(permuted[21:40])
```

```
[1] 0.5
```

Permutation test example (cont.)

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mean(permuted[1:20]) - mean(permuted[21:40])
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[1] 0.5
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# 2. replicate, returning difference of groups under null
reps = replicate(1e5, {perm=sample(c(control,treatment));mean(perm[1:20])-mean(perm[21:40])})
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reps = replicate(1e5, {perm=sample(c(control,treatment));mean(perm[1:20])-mean(perm[21:40])})
```

```
# 3. find p-value as appropriate tail
mean(reps >= mean(treatment)-mean(control))
```

```
[1] 0.04529
```

Test results

In the real world, for any given problem, the **null hypothesis is either true or false**. Depending on the result of your test, 4 overall outcomes are possible:

	H_0 rejected (positive/significant)	**H_0 not rejected (negative/insignificant)**
H_0 true	False positive (type I error)	True negative
H_0 false	True positive	False negative (type II error)

Note: you do NOT get to choose if H_0 is true or false, one of these is simply true for your reality; you only get to decide the outcome of your test (positive/negative).

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1. If H_0 actually true, probability of **type I error**, $\Pr[\text{reject } H_0 \mid H_0 \text{ true}] = \alpha$
   - This is the same α as when we set the **significance level** of a test
2. If H_0 actually false,
   - Probability of **type II error**, $\Pr[\text{do not reject } H_0 \mid H_0 \text{ false}] = \beta$
   - Probability of **true positive**, $\Pr[\text{reject } H_0 \mid H_0 \text{ false}] = 1 - \beta$, also called **power** or **sensitivity** of a test

Testing exercise

Suppose you are a quality assurance agent at a factory checking for defects. For each product, define H_0 as "there is no defect". You pass or fail products depending on if you think there's a defect or not. Label each of the following outcomes with the following labels: true positive, true negative, type I error, type II error.

a. You passed a defective product:

b. You passed a good product:

c. You failed a good product:

d. You failed a defective product:

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a. You passed a defective product: **type II error**

b. You passed a good product: **true negative**

c. You failed a good product: **type I error**

d. You failed a defective product: **true positive**

Testing exercise #2

The skin prick test (SPT) is a method for determining dog and cat allergies. A new SPT is developed and 200 random people from the population are tested. A patient is assumed to not have an allergy unless the test shows that they do. Their actual allergy status and the results of the test are below:

	Allergy	**No Allergy**
Positive SPT result	16	26
Negative SPT result	8	150

a. Estimate the type I error rate of SPT.

b. Estimate the type II error rate of SPT.

c. Estimate the power of SPT.

d. If the prevalence (allergy rate) in the population increases, will the power of the test increase, decrease, stay the same, or is there not enough information to tell? Explain!

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a. Estimate the type I error rate of SPT. **26/176**

b. Estimate the type II error rate of SPT. **8/24**

c. Estimate the power of SPT. **16/24**

d. If the prevalence (allergy rate) in the population increases, will the power of the test increase, decrease, stay the same, or is there not enough information to tell? Explain!
no change; power doesn't depend on prevalence

Rejection region

For a certain test, statistics that would give a positive result form the **rejection region**.

Example: Suppose a coin has true probability p of being heads (unknown to us). We wish to determine at $\alpha = 0.05$ if the coin is fair, so we flip it 200 times and count the number of heads X. (Exercise for home: plot the distribution of X assuming the coin is fair)

Let null: coin is fair ($p = 0.5$); for alternative consider each case, A: two-sided ($p \neq 0.5$), B: one-sided lesser ($p < 0.5$), C: one-sided greater ($p > 0.5$). Rejection region of each:

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```
# rejection region for A, # i.e. if X≤85 or X≥115 reject Ho
c(qbinom(0.025,200,0.5)-1, qbinom(0.975,200,0.5)+1)
```

```
[1]  85 115
```

```
# check both sides
c(2*pbinom(85,200,0.5), 2*pbinom(86,200,0.5))
```

```
[1] 0.04003719 0.05596574
```

```
c(2*(1-pbinom(115-1,200,0.5)), 2*(1-pbinom(114-1,200,0.5)))
```

```
[1] 0.04003719 0.05596574
```

```
# rejection region for B,
# i.e. if X≤87 we reject Ho
c(qbinom(0.05,200,0.5)-1)
```

```
[1] 87
```

```
# rejection region for C,
# i.e. if X≥113 we reject Ho
qbinom(0.95,200,0.5)+1
```

```
[1] 113
```

Rejection region (cont.)

Summary: for $X \sim \text{Binomial}(200, p)$, to test $H_0 : p = 0.5$, rejection region for each H_a:

- Case A: $H_a : p \neq 0.5$: region: $X \leq 85$ or $X \geq 115$
- Case B: $H_a : p < 0.5$: region: $X \leq 87$
- Case C: $H_a : p > 0.5$: region: $X \geq 113$

Rejection region (cont.)

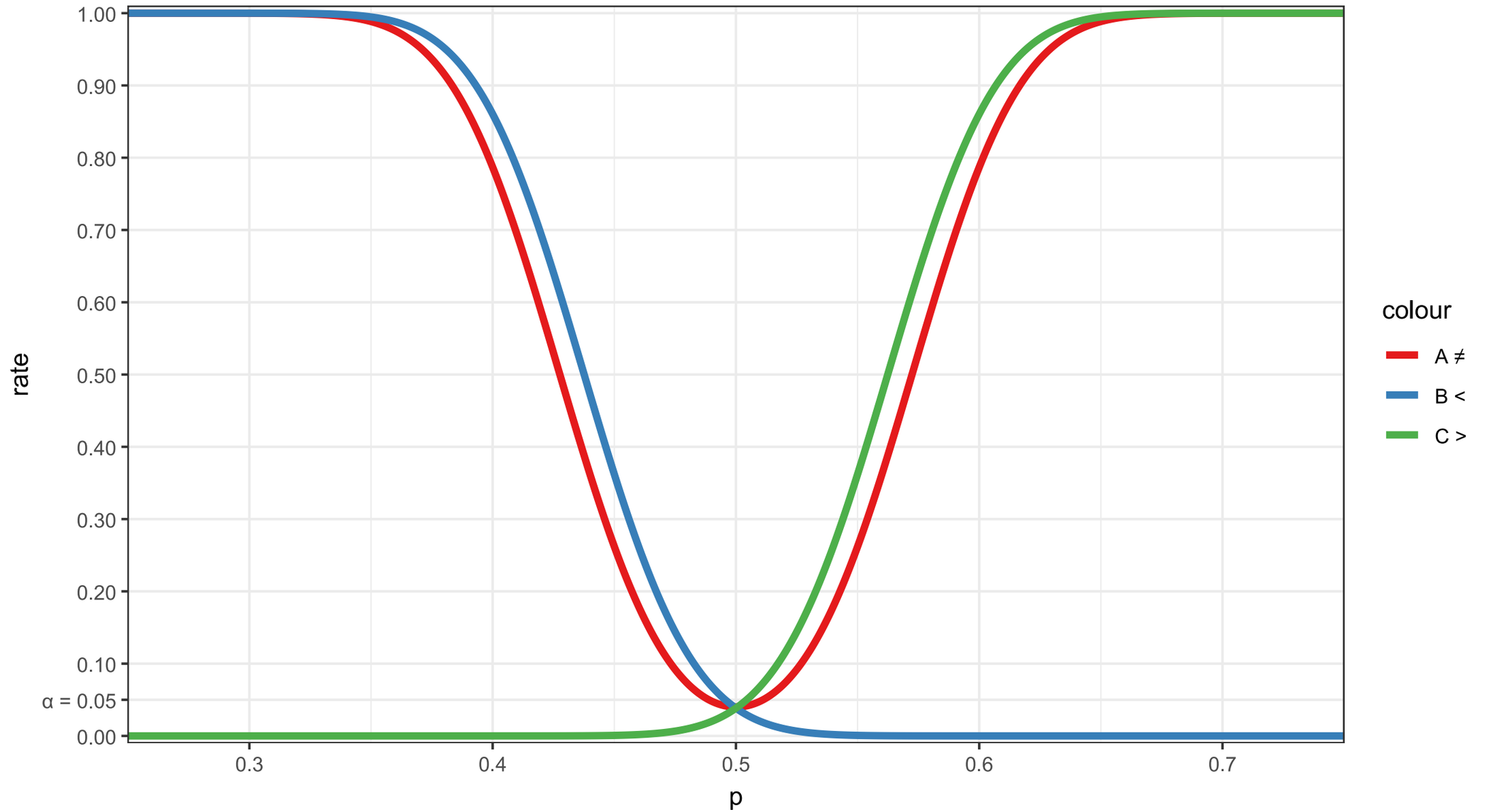
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- Case C: $H_a : p > 0.5$: region: $X \geq 113$

We can use this to find the **rejection rate** as a **function of** p, i.e. for coins with different levels of "fairness"/"unfairness" p, what's the **probability of a positive test**?

```
# define rejection rate functions for each case
fA = \(p) pbinom(85,200,p)+(1-pbinom(115-1,200,p))
fB = \(p) pbinom(87,200,p)
fC = \(p) 1-pbinom(113-1,200,p)
ggplot() + labs(x="p", y="rate") +
  geom_function(aes(color="A ≠"), fun=fA, xlim=c(.25,.75), n=1e4) +
  geom_function(aes(color="B <"), fun=fB, xlim=c(.25,.75), n=1e4) +
  geom_function(aes(color="C >"), fun=fC, xlim=c(.25,.75), n=1e4)
```

Rejection rate vs p for $\alpha = 0.05$



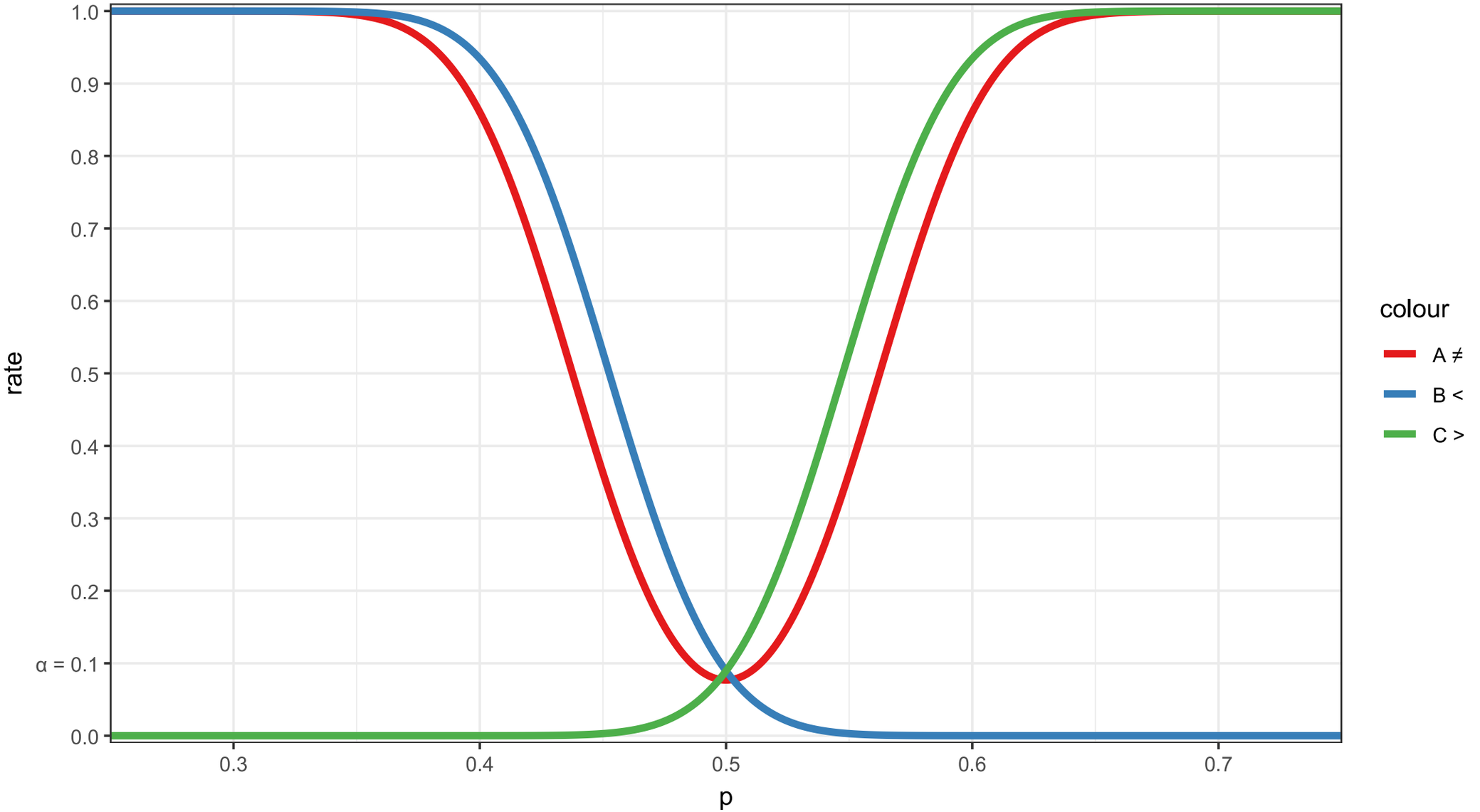
Rejection rate (cont.)

Key observations:

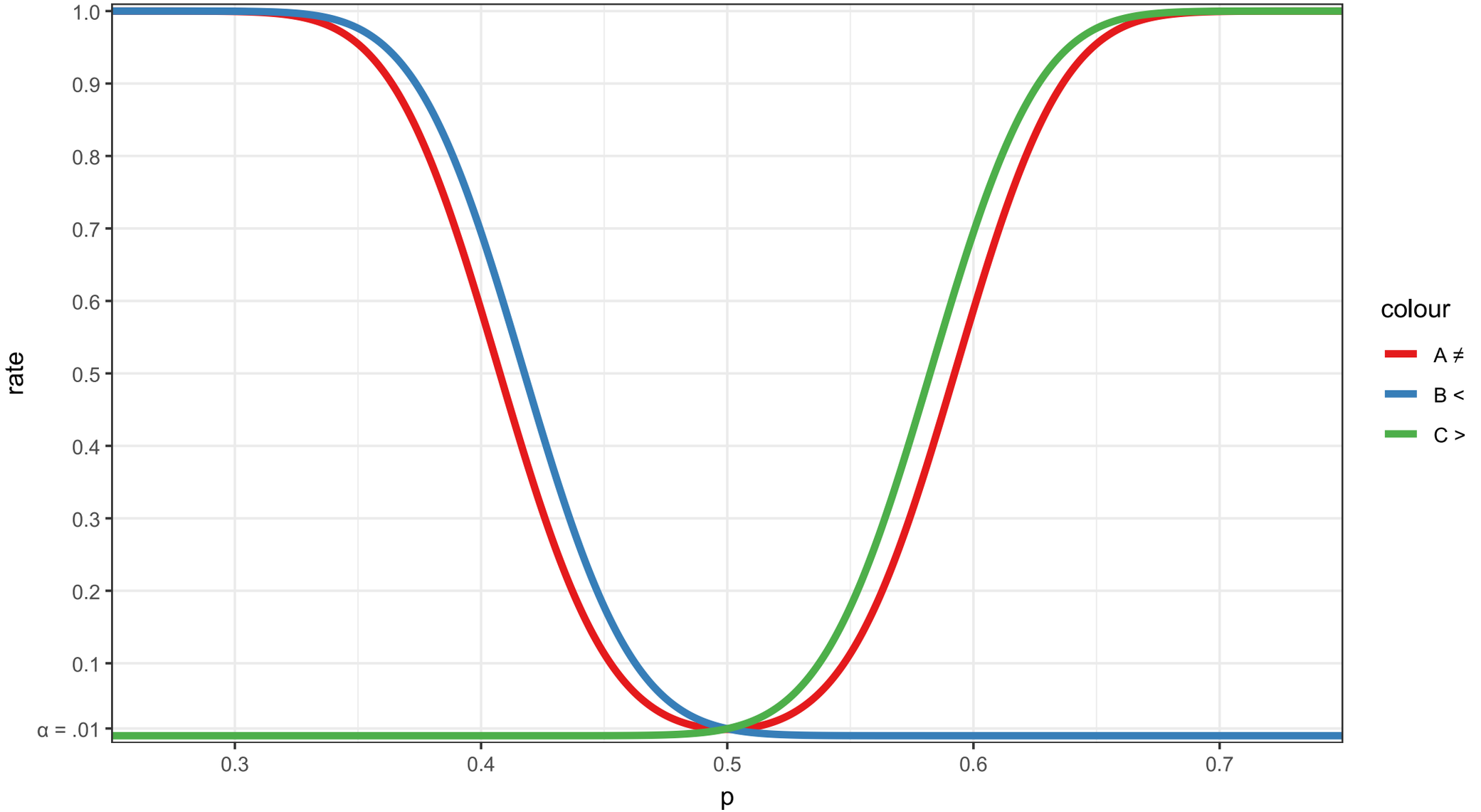
- When null is true, i.e. $p = 0.5$, rejection rate of $\alpha = 0.05$ approximately achieved
  - i.e. type I error rate is **correctly controlled**
- If not sure which side p is on, or both directions needed, two-sided test is better
  - i.e. non-zero **power** on both sides
- If side of p known, one-sided test is better
  - i.e. strictly **higher power** on corresponding side vs two-sided test
- Note: if wrong side is chosen, power quickly falls to 0!

How about for $\alpha = 0.10$ or 0.01?

Rejection rate vs p for $\alpha = 0.10$



Rejection rate vs p for $\alpha = 0.01$



How to choose statistics?

- Different statistics may be more or less appropriate for different problems, and even different hypotheses setup. For coin flips, some possibilities:
  - Number/proportion of total heads
  - Mean or max consecutive sequences of heads
  - Distribution of sequential pairs of heads
  - etc…

Remember: any quantity that tends to differ between H_0 and H_a can be used!

Choosing significance level α

- By default (from Fisher) $\alpha = 0.05$ just "felt right", but can choose any value
- Other choices may be more common in different fields, e.g.
  - In particle physics, "five-sigma" (i.e. one-tail p-value 5 SDs away on normal) is standard for discoveries, to ensure high confidence in new claims

```
# 5-sigma correspond to alpha ≈ 3e-7
pnorm(-5)
```

```
[1] 2.866516e-07
```

  - In social sciences, often difficult/costly to run experiments, so $\alpha = 0.10$ is common

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- By default (from Fisher) $\alpha = 0.05$ just "felt right", but can choose any value
- Other choices may be more common in different fields, e.g.
  - In particle physics, "five-sigma" (i.e. one-tail p-value 5 SDs away on normal) is standard for discoveries, to ensure high confidence in new claims

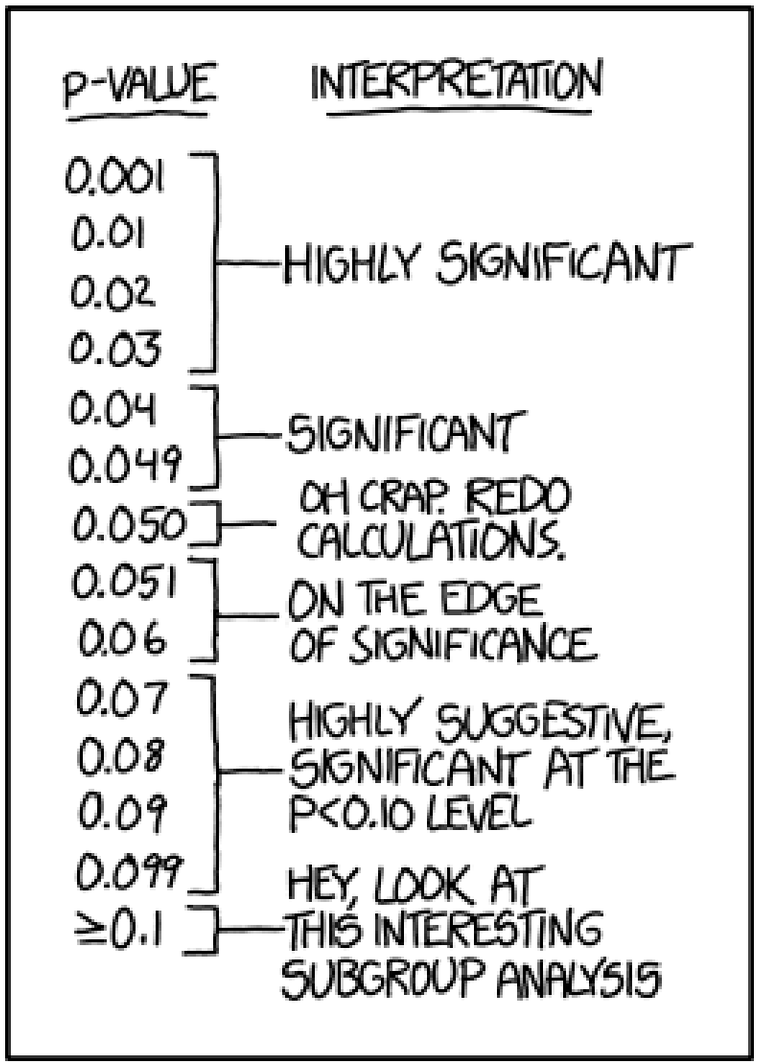
```
# 5-sigma correspond to alpha ≈ 3e-7
pnorm(-5)
```

```
[1] 2.866516e-07
```

  - In social sciences, often difficult/costly to run experiments, so $\alpha = 0.10$ is common

- Crucially, α should be set **before** experiment is run and NOT changed after analysis!
- These interpretation phrases, though somewhat vague, are commonly used:
  - $p > 0.10$ generally "not significant"
  - $p < 0.10$ often called "weakly significant"
  - $p < 0.05$ often called "significant"
  - $p < 0.01$ often called "strongly significant"

P-value interpretations



source: https://xkcd.com/1478/

Impact of different α values

Looking at table again:

	H_0 rejected (positive/significant)	**H_0 not rejected (negative/insignificant)**
H_0 true	False positive (prob $= \alpha$)	True negative (prob $= 1-\alpha$)
H_0 false	True positive (prob $= 1-\beta$)	False negative (prob $= \beta$)

- **Higher** α means **more rejections** (lower α means fewer rejections)
- Generally, **raising** α **lowers** β (lowering α raises β)
- Also, **raising** α **raises power** (lowering α lowers power)
- Generally, only way to lower both α, β is to increase sample size

Tradeoffs all around!